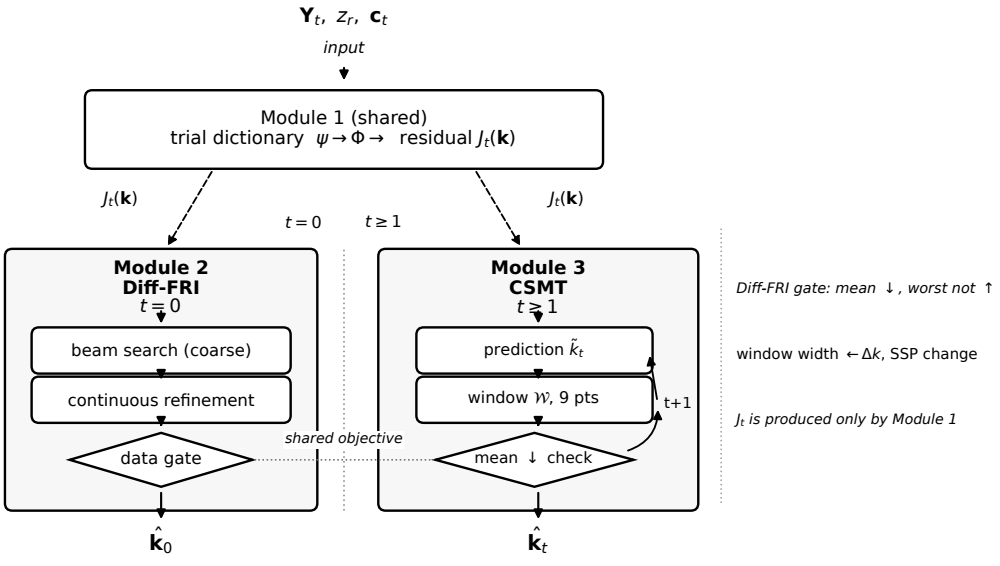
$\mathbf{Y}_t,\ z_r,\ \mathbf{c}_t$
input
Module 1 (shared)
trial dictionary $\psi \to \Phi \to$ residual $J_t(\mathbf{k})$
$J_t(\mathbf{k})$
$J_t(\mathbf{k})$
$t = 0$
$t \geq 1$
Module 2
Diff-FRI
$t = 0$
beam search (coarse)
continuous refinement
data gate
shared objective
Module 3
CSMT
$t \geq 1$
prediction $\tilde{k}_t$
window $\mathcal{W}$, 9 pts
mean ↓ check
t+1
$\hat{\mathbf{k}}_0$
$\hat{\mathbf{k}}_t$
Diff-FRI gate: mean ↓, worst not ↑
window width ← Δk, SSP change
J_t is produced only by Module 1